

# จงหาปริพันธ์ของข้อต่อไปนี

$$\int (x^4 + 3x^3 - 2x - 1) dx$$

$$\begin{aligned} \int f(x) dx &= \int x^4 + 3x^3 - 2x - 1 dx \\ &= \frac{x^{4+1}}{4+1} + \frac{3x^{3+1}}{3+1} - \frac{2x^{1+1}}{1+1} - 1x + C \\ &= \frac{x^5}{5} + \frac{3x^4}{4} - \frac{2x^2}{2} - x + C \\ &= \frac{x^5}{5} + \frac{3x^4}{4} - x^2 - x + C \end{aligned}$$

$$\int (x^2 + 2x) dx$$

$$\begin{aligned} \int f(x) dx &= \int (x^2 + 2x) dx \\ &= \frac{x^{2+1}}{2+1} + \frac{2x^{1+1}}{1+1} + C \\ &= \frac{x^3}{3} + \frac{2x^2}{2} + C \\ &= \frac{x^3}{3} + x^2 + C \end{aligned}$$

# จงหาปริพันธ์ของข้อต่อไปนี้

$$\int (x+5)^2 dx$$

$$\text{สูตร } (n+l)^2 = n^2 + 2nl + l^2$$

$$\begin{aligned} f(x) &= (x+5)^2 = x^2 + 10x + 25 \\ \int (x^2 + 10x + 25) dx &= \frac{x^{2+1}}{2+1} + \frac{10x^{1+1}}{1+1} + 25x + C \\ &= \frac{x^3}{3} + \frac{10x^2}{2} + 25x + C \\ &= \frac{x^3}{3} + 5x^2 + 25x + C \end{aligned}$$

$$\int \left( \frac{x^5 - 3x^3 - 4x - 7}{x^2} \right) dx$$

$$\begin{aligned} f(x) &= \frac{x^5 + 3x^3 - 4x - 7}{x^2} = \frac{x^5}{x^2} - \frac{3x^3}{x^2} - \frac{4x}{x^2} - \frac{7}{x^2} \\ &= x^3 - 3x - \frac{4}{x} - 7x^{-2} \\ \int (x^3 - 3x - \frac{4}{x} - 7x^{-2}) dx &= \frac{x^{3+1}}{3+1} - \frac{3x^{1+1}}{1+1} - 4 \ln|x| - \frac{7x^{-2+1}}{-2+1} + C \\ &= \frac{x^4}{4} - \frac{3x^2}{2} - 4 \ln|x| - \frac{7x^{-1}}{-1} + C \\ &= \frac{x^4}{4} - \frac{3x^2}{2} - 4 \ln|x| + \frac{7}{x} + C \end{aligned}$$

Class Activity

จงหาพื้นที่ล้อมรอบด้วยกราฟของ  $y = x + 15$ ,  $x = 1$  และ  $x = 6$

$$\begin{aligned} \int_1^6 (x + 15) dx &= \left[ \frac{x^2}{2} + 15x \right]_1^6 \\ &= \left( \frac{6^2}{2} + 15(6) \right) - \left( \frac{1^2}{2} + 15(1) \right) \\ &= \left( \frac{36}{2} + 90 \right) - \left( \frac{1}{2} + 15 \right) \\ &= \left( \frac{36}{2} + \frac{180}{2} \right) - \left( \frac{1}{2} + \frac{30}{2} \right) \\ &= \left( \frac{216}{2} \right) - \left( \frac{31}{2} \right) = \frac{185}{2} \text{ หน่วย} \end{aligned}$$